FPGA Hackathon 2026: Technical Report

Title:

Team Member Names and Email IDs:

Affiliation:

Application Domain:

1. **Abstract (in 250-500 words):** Problem statement, Proposed FPGA-based solution, Key architectural features, Major quantitative results
2. **Introduction:** Real-world problem description, Importance of Edge AI, Justification for FPGA-based implementation instead of cloud-based solutions, Related work and existing approaches, Motivation and objectives
3. **Novelty and Key Technical Contributions:** Novel AI/ML model approach (if any), Custom hardware accelerator architecture, Pipeline / parallel processing strategy, Resource optimization techniques, Hardware–software co-design approach (if used)
4. **Dataset Description:** Dataset source (with link), Number of samples, Input format (image size, signal length, feature dimensions), Train/test split
5. **AI/ML Model Description:** Model type (CNN, MLP, SVM, etc.), Architecture diagram, Layers / neurons, Activation functions, Input/output dimensions, Model parameter count
6. **Software Performance:** Accuracy, Precision / Recall / F1-score, CPU inference latency, GPU inference latency (if applicable), Comparison with existing software-based methods.
7. **Hardware Architecture:** Block diagram, Functional-level architecture, Dataflow description, Pipeline depth, Parallel computation units, Memory hierarchy, Area–time complexity analysis, Computational complexity. Quantizing technique used (Fixed-Point, Floating-Point, etc)
8. **RTL Implementation:** describe custom Verilog modules, interface design and implementation methodology (Vivado-flow).

   Note: Higher weightage will be given to custom, optimized RTL implementations over automatically generated or IP-based RTL designs.
9. **Functional Verification and Simulation:** Testbench description, Output waveforms, Functional correctness validation, Hw-Sw co-design Co-simulation (if used)
10. **FPGA Implementation Results (if available):**
    a. Experimental setup: FPGA board used (e.g., ZedBoard, Zybo, PYNQ) in experimental setup with photograph, Tool chain and Implementation methodology (e.g., Vivado RTL flow, Vivado–Vitis hardware–software co-design flow, Vivado–PYNQ flow, HLS4ML, System Generator), Clock configuration and target frequency, Communication interface used (AXI, UART, SPI, etc.), On-chip / soft processor usage (if applicable).
    b. Resource Utilization: LUT, FF, DSP, BRAM
    c. Performance Metrics: Clock frequency, Latency, Throughput, Power estimation, Memory utilization, Communication overhead (for HW–SW co-design)
    d. Comparative Performance Analysis: Pure software vs Pure hardware (FPGA) vs co-design (Fully accelerated and/or partially accelerated), vs CPU vs GPU; comparing with quantizing technique (Fixed-Point, Floating-Point, etc), Comparison with recent state-of-the-art works
    e. Scalability Discussion: How architecture scales for larger models, Multi-layer expansion, etc
11. **Conclusion:** Summary of contribution, Performance gains, Resource efficiency, Future improvements